

Quiz 1

Linear Algebra - 24DS

May 20, 2026

Time allowed: 50 Minutes

1. Think we are working in $\mathbb{R}^3$, can a set containing the zero vector be linear independent? Argue. **[10 minutes]**
2. Describe the all vectors formed by **linear combinations** a single nonzero vector. What geometric shape they are forming? Explain. **[10 minutes]**

3. Construct vectors $v_1 = \begin{bmatrix} a + b + c \\ -a + b + c \\ a - b - c \end{bmatrix}$, $v_2 = \begin{bmatrix} a + b - c \\ -a - b + c \\ -a - b + c \end{bmatrix}$, $v_3 = \begin{bmatrix} a - b + c \\ -a + b - c \\ -a - b - c \end{bmatrix}$

where a and b are last two digits from right of your student ID number.

For instance, enrollment number is **24DS78**, then $a = 7$ and $b = 8$

and $c = 7 + 8 = 15$. Check if vectors v_1, v_2 and v_3 are linear independent ? [30 minutes]